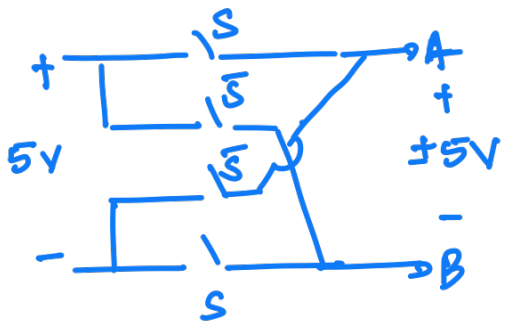
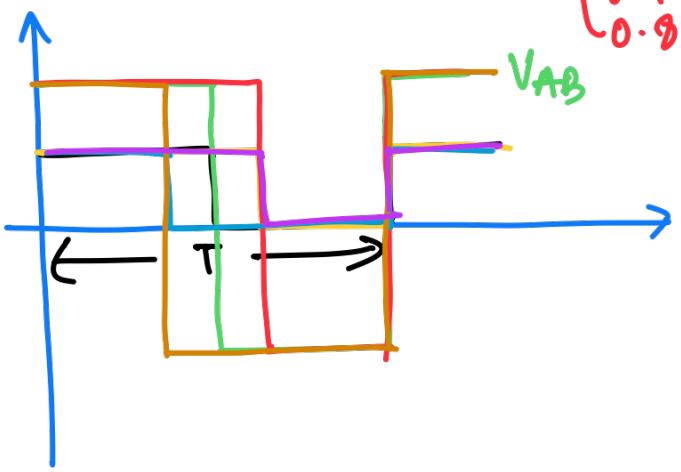
Lecture #23.

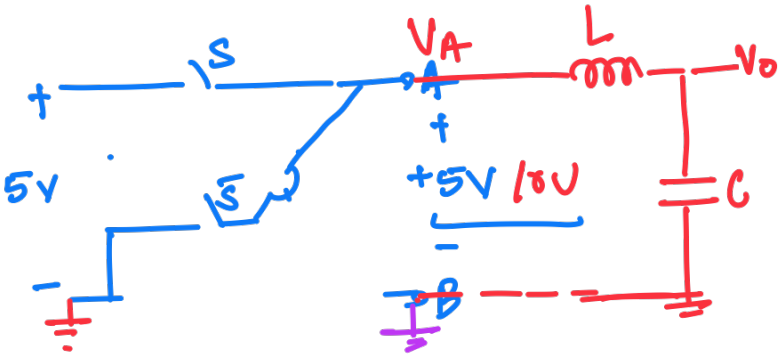


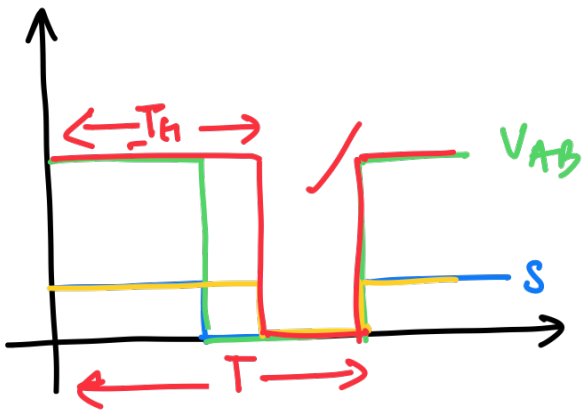
3.3 V
2.5 V
1.8 V
1.2 V
1.0 V
0.9
0.8

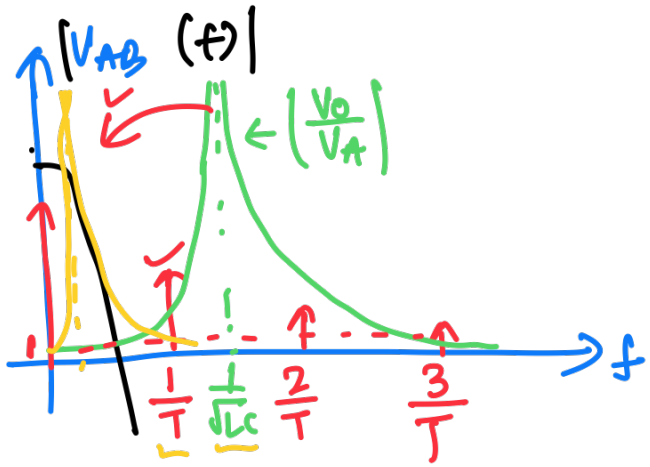
250 nm
180 nm
130 nm, 90 nm
65 nm
28 nm, 22
16 nm.



$$V_{AB} = a_0 + \sum_n a_n \cos(n\omega_s t) + b_n \sin(n\omega_s t)$$



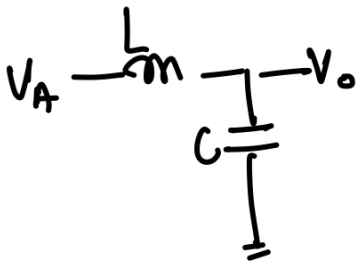


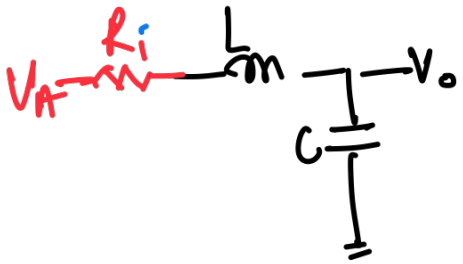


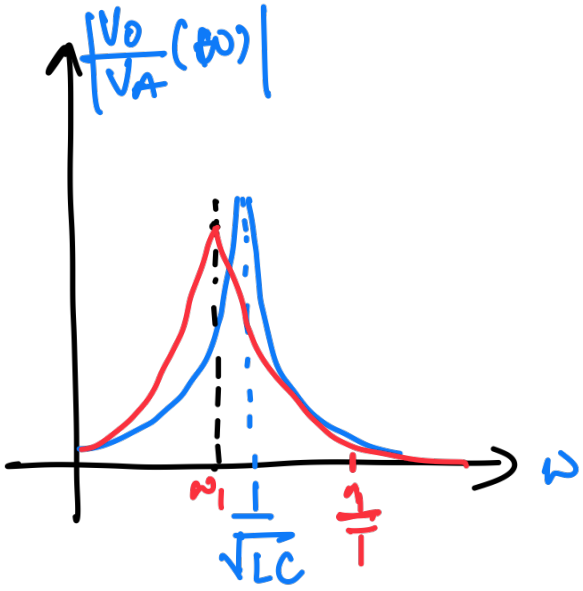
DC value, $a_0 = \frac{T_H}{T} \times VDD = \frac{VDD \times T_H + 0\,(T - T_H)}{T}$

$\frac{V_o}{V_A} = \frac{1}{1 + s^2 LC}$ at $\omega = \frac{1}{\sqrt{LC}}$ $\quad \left|\frac{V_o}{V_A}\right| = \infty$

$$\left|\frac{V_o}{V_A}(\omega=0)\right| = 1$$







$$\frac{V_o}{V_A} = \frac{V_{sc}}{\frac{1}{sC} + sL + R_1} = \frac{1}{s^2LC + sCR_1 + 1}$$

$$= \frac{1}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$

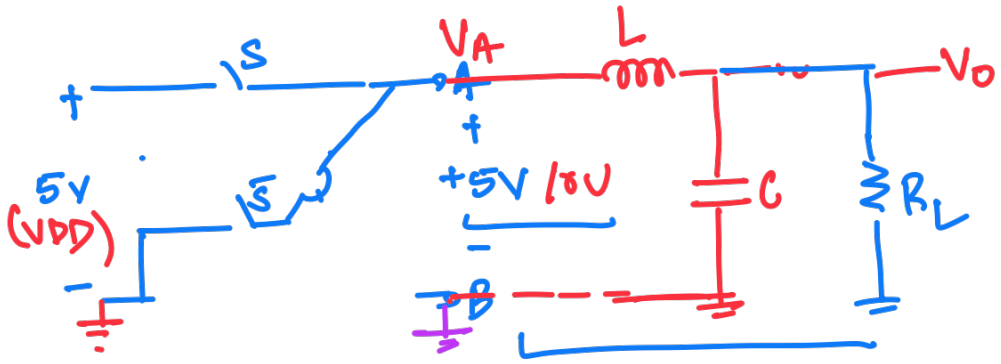
$$\omega_p = \frac{1}{\sqrt{LC}}$$

$$\omega_p Q_p = \frac{1}{CR_1}$$

$$Q_p = \frac{1}{R_1}\frac{1}{C}\sqrt{LC} = \frac{1}{R}\sqrt{\frac{L}{C}}$$

$$\left|\frac{V_o}{V_A}(j\omega)\right| = \left|\frac{1}{-\frac{\omega^2}{\omega_p^2} + j\frac{\omega}{\omega_p Q_p} + 1}\right|$$

$$= \frac{1}{\sqrt{\left(1-\frac{\omega^2}{\omega_p^2}\right)^2 + \left(\frac{\omega}{\omega_p Q_p}\right)^2}}$$



$$\frac{V_O}{V_A} = \frac{\left(R_L \| \frac{1}{sC}\right)}{\left(R_L \| \frac{1}{sC}\right) + sL} = \frac{R_L / (1+sCR_L)}{\frac{R_L}{1+sCR_L} + sL}$$

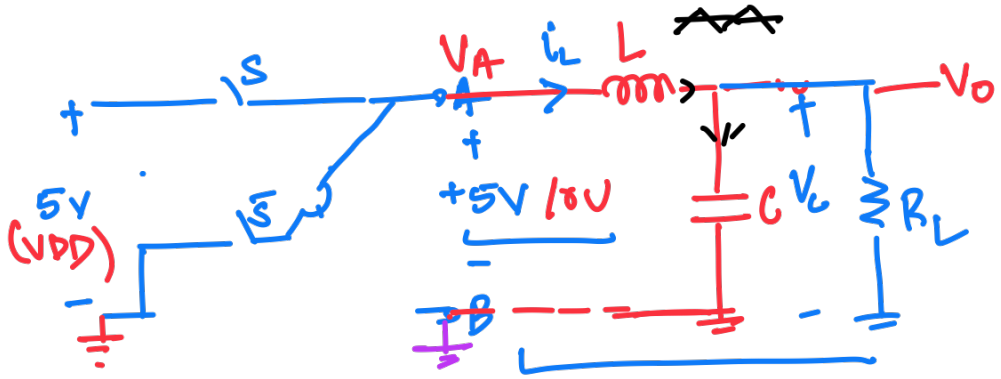
$$= \frac{R_L}{R_L + sL + s^2 LCR_L} = \frac{1}{1 + \frac{sL}{R_L} + s^2 LC}.$$

$$= \frac{1}{\frac{s^2}{\omega_p^2} + \frac{s}{\omega_p Q_p} + 1}$$

$$\omega_p = \frac{1}{\sqrt{LC}}$$

$$\omega_p Q_p = \frac{R_L}{L}$$

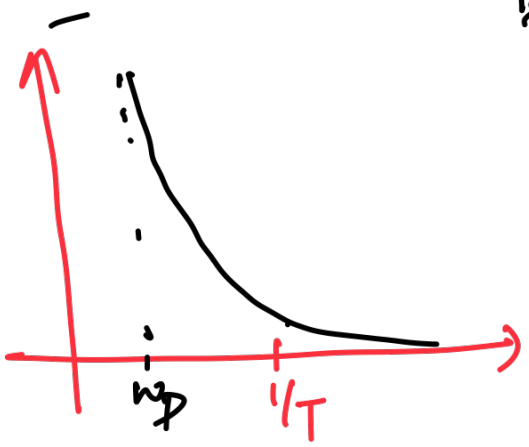
$$Q_p = \frac{R_L}{L}\sqrt{LC} = R_L\sqrt{\frac{C}{L}}$$

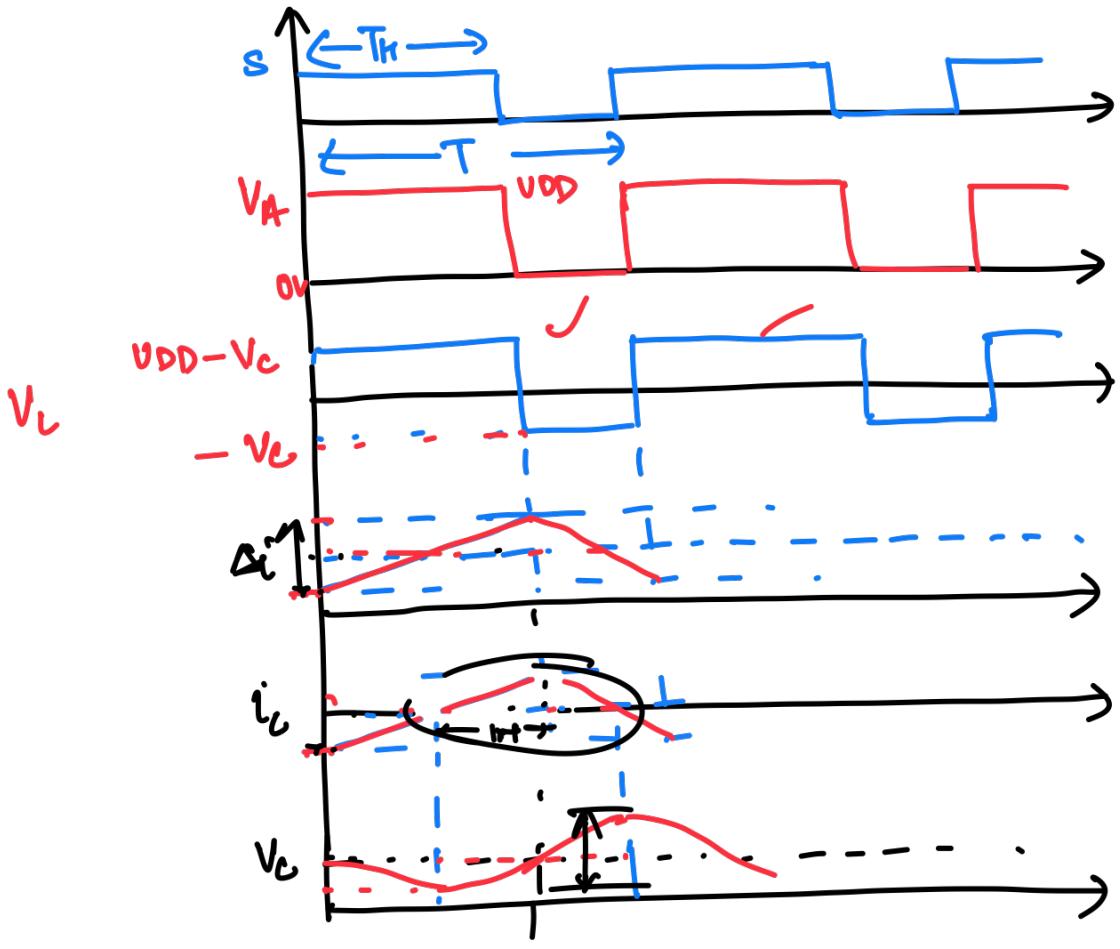


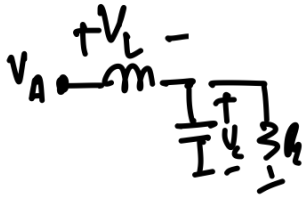
$$V_A = a_0 + \sum a_n \cos(n\omega_s t) + b_n \sin(n\omega_s t), \quad \omega_s = \frac{2\pi}{T_s}$$

$$\frac{V_o}{V_A} = \frac{1}{s^2 LC + s\frac{L}{R_L} + 1}$$

$$\left| \frac{V_o}{V_A}(\omega = 0) \right| = 1$$







$$V_L = V_A - V_c$$

$$= L\frac{di}{dt}$$

$$\int di = \int \frac{V_L}{L}\, dt$$

$$\Delta i = \frac{VDD - V_c}{L} \cdot T_H$$

$$= \frac{V_c}{L}(T - T_H)$$

$$V_c = \frac{T_H}{T} VDD$$

$$i = C\frac{dV}{dt}$$

$$\int dV = \int \frac{i}{C}\, dt$$

$$\Delta V_c = \frac{\Delta q}{C} = \frac{\frac{1}{2} \times \left(\frac{T_H}{2} + \frac{T_L}{2}\right) \times \frac{\Delta i_L}{2}}{C}$$

$$\Delta i = \frac{VDD - \frac{T_n}{T} VDD}{L} \cdot T_n$$

$$= \frac{VDD(1-D)}{L} \cdot \frac{T_n}{T} \cdot T$$

$$= \frac{VDD(1-D)D}{L \cdot f_s}$$

Duty Cycle $D = \frac{T_n}{T}$

Sampling freq. $f_s = \frac{1}{T}$

$\Delta i \propto \frac{f}{f_s}$

$\Delta i \propto \frac{1}{L}$

$$\Delta V_c = \frac{1}{8} \cdot \frac{T}{C} \cdot \frac{VDD(1-D)D}{L \cdot f_s}$$

$$= \frac{1}{8C} \frac{VDD(1-D)D}{L f_s^2}$$